DAY 4 | POST-READ

Architecture and MongoDB: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Frontend and backend	Frontend shows and collects; backend holds rules, data and secrets.
Middleware	Code that runs on every request: logging, auth, CORS.
API-based architecture	All clients use the same API contract, usually JSON over HTTP.
Matplotlib	plot, bar, hist, scatter, pie. Title, labels, legend. show() or savefig().
NoSQL and BSON	Flexible documents stored as Binary JSON with types such as ObjectId and dates.
CRUD	insert_one/many, find with filters, update with $set/$inc/$push, delete. Test filters first.
Indexes	create_index on fields you filter or sort by. Unique indexes stop duplicates.
Embed vs reference	Embed what you read together; reference what is shared or grows without limit.

Part B: Going beyond today

1. Subplots: several charts in one figure

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
ax1.bar(cats, totals)
ax2.plot(months, sales)
fig.tight_layout()
```

2. Upserts

If no document matches, upsert=True inserts one. Useful for syncing data.

```
tickets.update_one({"ticket_no": "T-100"}, {"$set": {"status": "open"}}, upsert=True)
```

3. Aggregation preview

The aggregation pipeline groups and counts inside the database, instead of in Python. You will use it on Day 7 for the SmartDesk analytics endpoint.

```
tickets.aggregate([{"$group": {"_id": "$priority", "n": {"$sum": 1}}}])
```

4. Checking an index is used

On a real MongoDB, find(...).explain() shows whether a query used an index (IXSCAN) or scanned the whole collection (COLLSCAN). mongomock does not support explain, so try this on Atlas.

5. Connection strings and secrets

An Atlas connection string contains a password. Never hard-code it or push it to GitHub. Read it from an environment variable: os.environ["MONGO_URI"].

6. Stateless servers

Good APIs do not remember anything between requests; every request carries what it needs. This lets you run many copies of a server behind a load balancer.

Part C: Check yourself

- What is the difference between frontend, backend and middleware?
- What does BSON add that JSON does not have?
- What is the difference between $set and $inc?
- Why does every extra index slow down writes a little?
- Would you embed or reference a ticket's customer? Why?

Part D: Practice ideas for tonight

- Create a free MongoDB Atlas cluster and connect to it using the Option B cell.
- Design documents for a movie app: movies, reviews and users. Decide what to embed and what to reference.
- Chart SmartDesk tickets per day for a week.

Coming up on Day 5: REST, HTTP and FastAPI, then Basics assessment 1 (Python, MongoDB, FastAPI core).

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today